



The Parameters To Evaluate Autonomous Vehicles

An autonomous vehicle (AV) relies on devices and components that exchange, share and interpret data. Interoperability of these devices and components ensures that these can form an integrated ecosystem within the vehicle, seamlessly communicating with each other without sacrificing performance



(Credit: <https://medium.com>)

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Autonomous vehicle (AV) testing and validation are vital for making AVs safe for humans on the road. Traditional prototype-based testing takes time and large budgets, so engineers need to take advantage of physics-based and virtual reality (VR) simulations. These digital tools will enable engineers to overcome the challenges of testing and validating AV designs.

As traffic can be erratic, AVs will encounter unique scenarios every day, like a car running on the wrong side on the road. Testing will check the AVs' reac-

tion to such unique scenarios. On-road prototypes would rarely experience these conditions, and setting up such scenarios would be dangerous and costly. Instead, testing and validating AVs for such scenarios can be done using simulation for fast and cost-effective results.

Vehicles controlled by artificial intelligence (AI) need to be tested for every conceivable iteration of traffic, conditions and signage. As physically testing these scenarios can be risky to pedestrians and other vehicles, engineers can explore the AV domain of validity using parameterisation, simulation and VR. This will help run multiple permutations in a fraction of the time for different stop-sign scenarios automatically.

AV testing parameters

AVs need a wide array of sensors to get a clear picture of the world around them, including cameras, lidars, radars, thermal cameras, GPSes and more. Glare can affect various sensors and, hence, these are tested and validated under various conditions using simulation where one, or multiple, sensors fail. Simulations can help model faulty sensors to determine how the AVs will handle such situations.

"Safety standards for AVs are absolutely critical for public acceptance of the new technology," says Greg McGuire, associate director of M-city autonomous vehicle testing lab at University of Michigan. "Without them, it is hard to decide whether self-driving cars are safe, and to gain public trust.